

# ADITYA PRASANNA MOGARE

✉ [mogare@usc.edu](mailto:mogare@usc.edu)

☎ +1 (213) 756-4828

🌐 [LinkedIn](#)

🌐 [Portfolio](#)

🌐 [GitHub](#)

📄 [Publication](#)

## EDUCATION

- **University of Southern California** Los Angeles, CA  
Master's of Science - Computer Science; **GPA: 3.70/4.0** Aug. 2024 – May 2026
- **University of Mumbai** Mumbai, India  
Bachelor's of Technology -Computer Engineering; **GPA: 9.63/10** Aug. 2020 – May 2024

## WORK EXPERIENCE

- **INTERDEPENDENT Studios** Los Angeles, CA  
Software Engineering and AI Automation Intern May 2025 – Aug. 2025
  - **Engineered** neural network and transformer models to predict film budgets, **securing \$10M** in funding and **driving 2 multimillion-dollar** greenlight decisions.
  - **Automated** a multimodal ML pipeline via **Flask** and **LLM APIs** to process **5,000+** **screenplays**, training an NLP model that predicted IMDb scores with 97% accuracy
  - **Deployed** scalable microservices via **Dockerized** CI/CD pipelines, reducing build-test-deploy time by over 60% from 2 weeks to 5 days.
  - **Architected** real-time distributed collaboration features using **React**, **Flask**, and the **Miro SDK**, implementing idempotent webhooks to improve sync reliability by **45%**.
- **University of Southern California** Los Angeles, CA  
Machine Learning Research Intern Jan. 2025 – Jun. 2025
  - **Designed** PyTorch CNNs (**ResNet50**) for **OCTA**-based diabetic retinopathy detection, utilizing **CLAHE preprocessing** and real-time augmentation to achieve **85%** validation accuracy.
  - **Enhanced** rare disease classification by mitigating dataset class imbalance, leveraging dynamic class weighting and strategic downsampling to boost minority-class recall by **30%**.
  - **Streamlined** machine learning pipelines by containerizing training and inference on **AWS EC2**, improving **ROC-AUC** by **5%** and reducing overall training time via efficient batch scheduling
- **KansasTek** Kansas, USA (Remote)  
Software Engineering and AI Intern Aug. 2022 – Aug 2023
  - **Built** a GPT-3 question-generation microservice achieving **sub-200 ms** response times by implementing **Flask APIs** and **MongoDB caching layers**, accelerating the candidate interview pipeline.
  - **Optimized** MongoDB aggregation pipelines to reduce data validation errors by **25%** by refactoring legacy backend code to **SOLID principles**, significantly improving system scalability
  - **Spearheaded** React.js and TypeScript frontend UI development across **4 core AIRecruiter** components by integrating Jest end-to-end testing, guaranteeing cross-module stability and improving administrator usability.

## PROJECTS

- **Lucerna - Defense Intelligence Platform (DoD)** **Github Link**  
React, Django, PostgreSQL, Redis, Celery, Docker, LLMs, Text-to-SQL Jan 2026- Present
  - **Developed** a secure, enterprise-grade Data Intelligence Platform for the U.S. Air Force (28th Contracting Squadron, Ellsworth AFB) via the **USC Hacking for Defense (H4D)** program, eliminating a critical procurement visibility gap across siloed legacy ERP systems
  - **Translated** natural language analytics queries into optimized SQL via a **Text-to-SQL intelligence engine**, executing against an OLAP database and passing results through a hosted LLM to auto-flag high-risk and expiring U.S. Air Force contracts
  - **Established** end-to-end ETL pipelines supporting **multi-format ingestion** (Excel uploads, live Google Sheets sync) into dynamically generated PostgreSQL tables — **reducing manual data-entry hours** for DoD contracting officers
- **LurisQA - Legal AI Assistant** **Github Link**  
Python, Transformers, LoRA, Sentence Transformers, FAISS, RAG Jan. 2025 – Jun 2025
  - **Fine-tuned** open-source LLMs (**LLaMA**, **Mistral**) reducing GPU memory consumption by **4x** utilizing **LoRA**, enabling scalable and cost-effective model deployment for legal question-answering
  - **Constructed** end-to-end ingestion pipelines indexing **100,000+** **U.S. legal documents** via SentenceTransformers and FAISS, achieving sub-second semantic retrieval for an advanced Legal QA system
- **Automaton – Distributed Workflow Automation Platform** **Github Link**  
React, Typescript ,Node.js, Express.js ,OpenAI , Socket.io, npmworkspaces Jul. 2025 – Present
  - **Engineered** a cloud-native workflow automation platform using **Node.js** and **React Flow**, reducing manual task configuration time by **40%** through a drag-and-drop visual designer.
  - **Integrated OpenAI** and **Socket.IO** to enable real-time, AI-enhanced workflow execution, increasing user task-completion speed by **35%** through automated node generation

## PUBLICATIONS

- **Building a Virtual Reality-Based Framework for the Education of Autistic Kids** – *Wiley-Scrivener* May. 2023

## TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, SQL, C,C++, Java
- **AI/ML:** OpenAI Agents SDK, LLaMA, Mistral, LoRA, RAG, Hugging Face, PyTorch, FAISS, RAG, LiteLLM, CNN.
- **Web & Databases:** React.js, Next.js, Flask, Django, Node.js, Express.js, PostgreSQL, MongoDB, DynamoDB, Redis.
- **Cloud & DevOps:** AWS (EC2), Docker, Kubernetes, Jenkins CI/CD, Microservices, Event-Driven Architecture.